

Your grade: 100%

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Next item →

1. Based on this lesson, what is the best way to describe a Kalman filter?

1 / 1 point

- ☐ As an electronic device incorporating microcontrollers and transistors.
- ☒ As an algorithm usually implemented as a computer program, designed to estimate the state of a dynamic system.
- ☐ As an analog circuit using operational amplifiers and passive components to filter noise out of a signal.
- ☐ As a PC-based program incorporating a graphical user interface and internet communications.



Correct
Yes, this is the purpose and typical way to implement a Kalman filter.

2. Based on this lesson, which of the following applications might be based on a Kalman filter? (Select all that apply.)

1 / 1 point

☒ A state estimator



Correct
Yes, state-estimation algorithms often use Kalman filters. You will learn more about this in the fourth week of this course.

☒ A target-tracking system



Correct
Yes, target-tracking systems often use Kalman filters. You will learn more about this in the second course in the specialization.

☒ A navigation system



Correct
Yes, navigation systems often use Kalman filters. You will learn more about this in the fourth course in the specialization.

☒ A parameter estimator



Correct
Yes, parameter-estimation algorithms often use Kalman filters. You will learn more about this in the third course in the specialization.

3. According to this lesson, what information or data do we need to have to be able to design and implement a Kalman filter? (Select all that apply.)

1 / 1 point

- ☐ An accurate to-scale physical replica (model) of the system whose state we seek to estimate.
- ☐ Design plans for the system whose state we seek to estimate.
- ☒ Sensor measurements from the system whose state we seek to estimate.



Correct
Yes. The measurements are observations of system behavior that are indirectly or directly related to the state we seek to estimate and are used by the Kalman filter to make the state estimate.

☒ A set of math equations describing the system whose state we seek to estimate.



Correct
Yes. These equations define a "mathematical model" that describes how the measurements are related to the internal state.